

This course can change your life (if you let it)

- permeates all of pure math
- applications to science, engineering, econ, ...
- [linear algebra + calculus + statistics + programming]

Study of linearity recognize and use linearity (even when it is hidden)

- linear equations are simplest
 - Even when situation is not linear, we can use linearity to understand it e.g. use tangent line to approximate a function
- "If you wanna start thinking, first learn how to think straight" ([stackexchange: why-study-linear-algebra](#))

What the course will involve

- linear algebra (calculations!) AND geometry, see [3blue1brown](#) "Essence of linear algebra" playlist on youtube: preview (5min), chapter 1 (vectors-10min)
- writing proofs (absolutely crucial at CMI)

Eugenia Cheng "How to write proofs: a quick guide"

- Practical
- Text: Artin (default), Strang (concrete), Lay
 - Halmos / Hoffman-Kunze (more abstract)
 - Class videos + notes will be posted (link on moodle)
 - Weekly tutorials in small groups

Evaluation: 40-50% final, 25-30% midterm, 10-20% quizzes
10-20% HW, MOST IMPORTANT PART OF THE COURSE

Generally due on Tue by 1pm on moodle

TeX or neatly written with full reasoning

Linear algebra is about _____

Several answers

- vectors and matrices

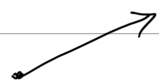
- solving systems of linear equations

- vector spaces and linear maps

[Thinking without a fixed coordinate system]

What is a vector?

- physics: magnitude and direction (in 2-d, 3-d)



starting point doesn't matter

Very useful, but for now put this out of mind

- Using coordinates $p\hat{i} + q\hat{j}$ = vector joining origin to the point (p, q)

For us, for now, a vector in $\mathbb{R}^n$ is just a tuple of n real numbers. By convention we'll write it as a column

$\mathbb{R}^n = \left\{ \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} \mid v_i \in \mathbb{R} \right\}$ = set of vectors in n -dimensional space

Notation: $\bar{v}, \bar{w}, \bar{a}, \bar{b}$ (will drop bars/arrows soon)

Operations on vectors

$$\mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

① addition: slotwise

$$\begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} w_1 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{bmatrix}$$

properties: For all vectors v, w, u

$\mathbb{R}^n$ forms an abelian group under $+$

$$\left\{ \begin{array}{l} \cdot \quad \bar{v} + \bar{w} = \bar{w} + \bar{v} \\ \cdot \quad (\bar{v} + \bar{w}) + \bar{u} = \bar{v} + (\bar{w} + \bar{u}) \\ \cdot \quad \text{There is a } \bar{0} \text{ vector, namely } \bar{0} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \\ \cdot \quad \bar{v} + \bar{0} = \bar{v} \\ \cdot \quad \text{Negative of } \bar{v} \text{ exists: } -\bar{v} := \begin{bmatrix} -v_1 \\ \vdots \\ -v_n \end{bmatrix} \text{ satisfies} \\ \quad \bar{v} + (-\bar{v}) = \bar{0} \end{array} \right.$$

② Scalar multiplication:

$$\mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

$$c \cdot \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix} = \begin{bmatrix} cv_1 \\ \vdots \\ cv_n \end{bmatrix}$$

properties

- $1 \cdot \bar{v} = \bar{v}$
- $(c_1 c_2) \cdot \bar{v} = c_1 \cdot (c_2 \cdot \bar{v})$
- $(c_1 + c_2) \cdot \bar{v} = (c_1 \cdot \bar{v}) + (c_2 \cdot \bar{v})$
- $c \cdot (\bar{v} + \bar{w}) = (c \cdot \bar{v}) + (c \cdot \bar{w})$

Defⁿ A vector space over real numbers is a set V along with two operations

$$\left. \begin{array}{l} V \times V \xrightarrow{+} V \\ \mathbb{R} \times V \xrightarrow{\cdot} V \end{array} \right\} \text{ satisfying the properties listed above}$$

Matrices: rectangular array of numbers

$r \times c$ matrix has r rows & c columns

• We can multiply $r \times c$ matrix with a column vector in $\mathbb{R}^c$

• More generally we can multiply matrices provided the sizes are appropriate

$$\left. \begin{array}{l} A \text{ } p \times q \text{ matrix} \\ B \text{ } q \times r \text{ matrix} \end{array} \right\} \Rightarrow AB \text{ is } p \times r \text{ matrix}$$

BA need not be defined!

(will be defined only if $p=r$)

Why is the definition the way it is? *will see a conceptual answer later on.*

• Can add matrices of the same size: slotwise

properties of matrix multiplication: Cheek!

$$(AB)C = A(BC)$$

$$(A+B)C = AC + BC$$

$$C(A+B) = CA + CB$$

I = identity matrix

$$= \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$$

$$A \text{ } p \times q \left\{ \begin{array}{l} A I_{q \times q} = A \\ I_{p \times p} A = A \end{array} \right.$$

Exercise: Compact definition of matrix multiplication using matrix units & assuming distributivity

E_{ij} = matrix with $\begin{cases} 1 \text{ in row } i \text{ \& column } j \\ 0 \text{ everywhere else} \end{cases}$

(Ex) $E_{ij} E_{kl} = ?$

A System of linear equations can be written in terms of a matrix and a vector

$$\begin{cases} 2x + 3y = 7 \\ 5x + 8y = 9 \end{cases} \Leftrightarrow \begin{bmatrix} 2 & 3 \\ 5 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 7 \\ 9 \end{bmatrix}$$

In general if I have a system of r equations in c variables, I can write it as

$$A\bar{x} = \bar{b} \quad \text{where} \quad \begin{cases} A \text{ is } r \times c \text{ matrix} \\ \bar{x} \text{ is an unknown vector in } \mathbb{R}^c \\ \bar{b} \text{ is a vector in } \mathbb{R}^r \end{cases}$$

Think about the geometry of solving

Discussed in class {
• one eqⁿ in 2 variables
• two eqⁿ in 2 variables

Discussed → • one/two/three eqⁿ in 3 variables
after class

} Each eqⁿ linear and not contradictory (i.e. not of type $0x + 0y = 1$)

[Tomorrow]: we will see a systematic way to solve an arbitrary linear system $Ax = b$. In particular

- how to detect inconsistency
- how to count number of solutions (degrees of freedom)
- how to write all solutions